RP-200 Centrifugal Pump Manual

Overview

The RP-200 is a single-stage centrifugal pump that transfers product from the blend tank to the filler bowl on the Riverbend Packaging filling line. It is driven by a 5 hp motor through a close-coupled shaft and uses a single mechanical seal with a flush port. This manual covers normal operation, troubleshooting, and the preventive maintenance (PM) schedule.

Specifications

Item	Value
Model	RP-200
Rated flow	60 gpm at 40 ft head
Motor	5 hp, 460 V, 3-phase, 1750 rpm
Seal	Single mechanical seal, carbon/ceramic, EPDM elastomers
Max inlet pressure	30 psi
Max product temperature	140 F
Discharge size	2 in. tri-clamp

Operation

Startup

1. Confirm the blend tank is above the low-level mark. The pump must never run dry.
2. Open the suction valve fully, then crack the discharge valve one quarter turn.
3. Check that the seal flush line is open and the sight glass shows flow.
4. Press START at the local panel and confirm rotation matches the arrow on the casing.
5. Open the discharge valve slowly while watching the discharge pressure gauge. Normal operating pressure is 18 to 22 psi.

Shutdown

1. Close the discharge valve slowly.
2. Press STOP and wait for the shaft to stop.
3. Close the suction valve if the pump will be idle for more than one shift.
4. Flush the pump with potable water if product will not be restarted within two hours.

WARNING: Never run the RP-200 against a closed discharge valve for more than 30 seconds. Heat build-up damages the seal and can flash product to steam.

Troubleshooting

Symptom	Likely cause	Corrective action
Seal leak at the shaft	Worn seal faces or dry running	Replace mechanical seal; check flush flow
Leak at casing clamp	Damaged gasket	Replace tri-clamp gasket
Low discharge pressure	Air in suction line or worn impeller	Re-prime pump; inspect impeller clearance
Pump noise or rattling	Cavitation or bearing wear	Check suction level; inspect motor bearings
Motor trips on overload	Closed valve or rubbing impeller	Check valve positions; inspect impeller
Product foaming	Air leak at suction fitting	Tighten fittings; replace suction gasket
Excess vibration	Misalignment or impeller imbalance	Check coupling; balance impeller

Seal Leak

Sanitizer or product leaking at the shaft indicates seal wear. Small weeping of a few drops per minute can continue for a limited time, but a steady stream requires immediate shutdown. To replace the seal:

1 Stop the pump and perform lockout/tagout on disconnect MD-6 and close product valves PV-1 and PV-2.

2 Drain the casing into a container and rinse residue.

3 Remove the casing clamp and slide the casing off the impeller.

4 Remove the impeller nut and impeller. Pull the old seal assembly from the shaft.

5 Clean the shaft and seal seat. Inspect for scoring.

6 Lubricate the new seal elastomers with food-grade lubricant only. Never touch the seal faces with bare fingers.

7 Reassemble in reverse order and torque the impeller nut to 35 ft-lb.

8 Remove locks and restore service. Run for five minutes and check for leaks.

Any chemical released from the seal must be handled under the Chemical Spill and Sanitizer Handling procedure, and product exposed to the leak must be reported to Quality.

Preventive Maintenance Schedule

Interval	Task	Duration
Daily	Visual check for leaks, noise, and vibration	5 min
Weekly	Check seal flush flow and pressure gauge accuracy	10 min
Monthly	Inspect gaskets and tri-clamps; verify coupling guard	20 min
Quarterly	Lubricate motor bearings (2 pumps of grease)	15 min
Annually	Inspect impeller, replace mechanical seal, megger motor	3 hr

Spare Parts

Keep on hand: mechanical seal kit RP200-SK, tri-clamp gaskets (EPDM), impeller nut, and motor bearing set. Order lead time for the impeller is three weeks.

Records

Record each PM task in the maintenance log with date, technician, and findings. Any seal replacement is noted with the run hours since the last replacement to support life-cycle tracking.